CASUS
CENTER FOR ADVANCED
SYSTEMS UNDERSTANDING

ANIMOVE 2025

Population-level inferences

Using the 'ctmm' R package

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MAX PLANCK INSTITUTE OF MOLECULAR CELL BIOLOGY AND GENETICS

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SACHSEN

Diese Maßnahme wird mitfinanziert mit Steuermitteln auf Grundlage des vom Sächsischen Landtag beschlossenen Haushaltes

Analyses of ecological data should always account for the ***uncertainty*** in the process(es) that generated the data.

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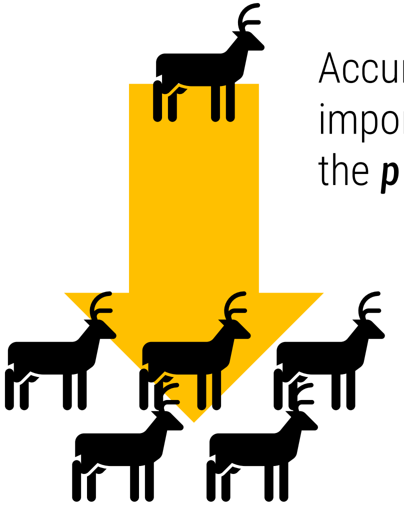
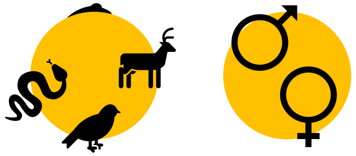


Accurately estimating area requirements is of utmost importance for conservation, from the ***individual*** to the ***population level***.



We want to quantify the effect of covariates, such as ***species***, ***sex***, ***body size***, ***age***, ***habitat***, ***anthropogenic impact***, *etc*…

…even if we are comparing different populations with different ***movement behaviors*** or ***sampling schedules***.

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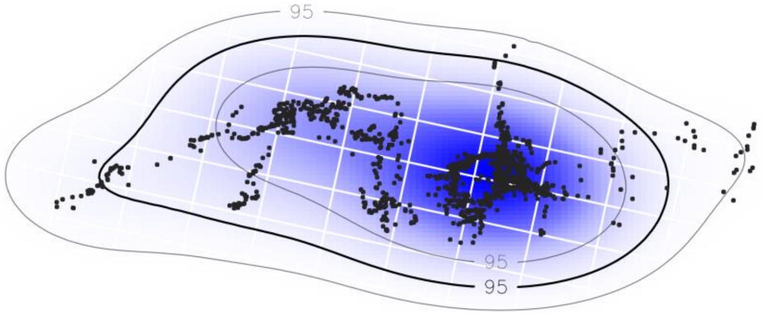
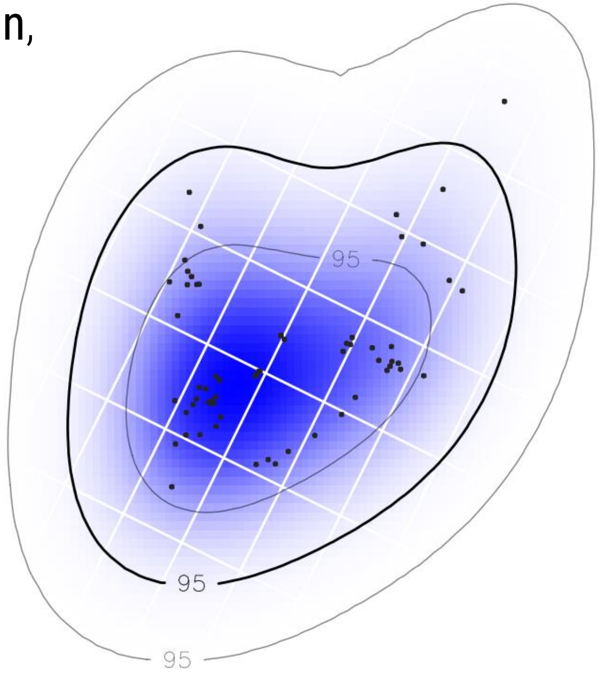
Conventionally,

Taking KDE and MCP home-ranged estimates and feed them into general purpose statistical analyses, which assumes that the individual home-range areas are measured exactly

e.g., a single population would be described by its **sample mean**, and two populations would be compared with a ***t*-test**





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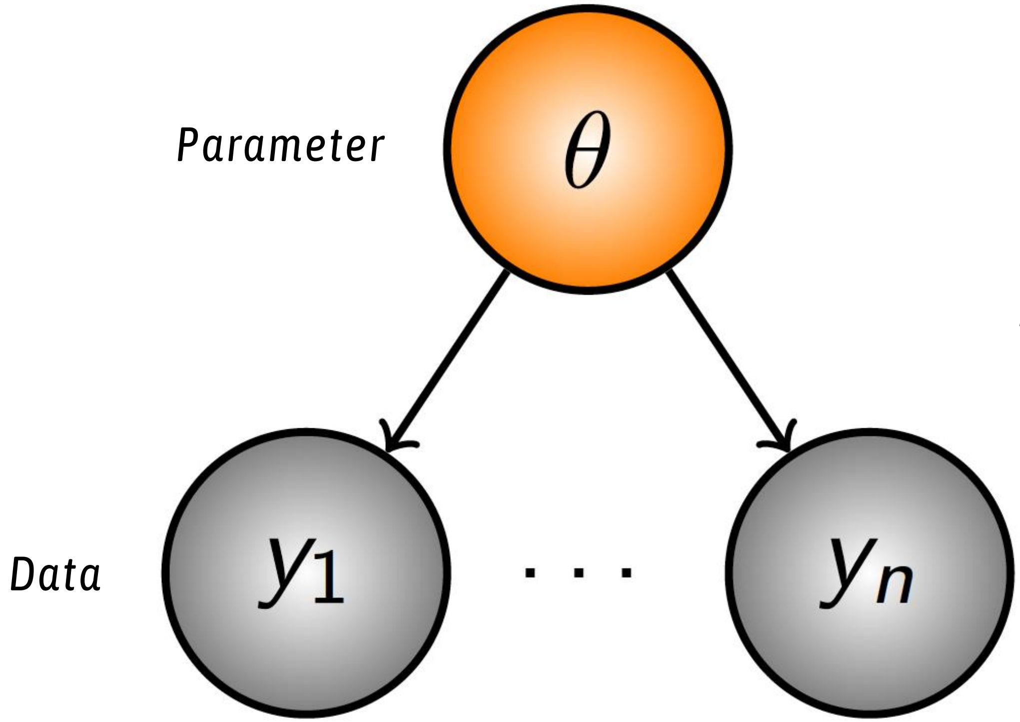
NON-HIERARCHICAL MODELS

How does data inform parameters?



"Complete **pooling**"

Adapted from Midway (2008)

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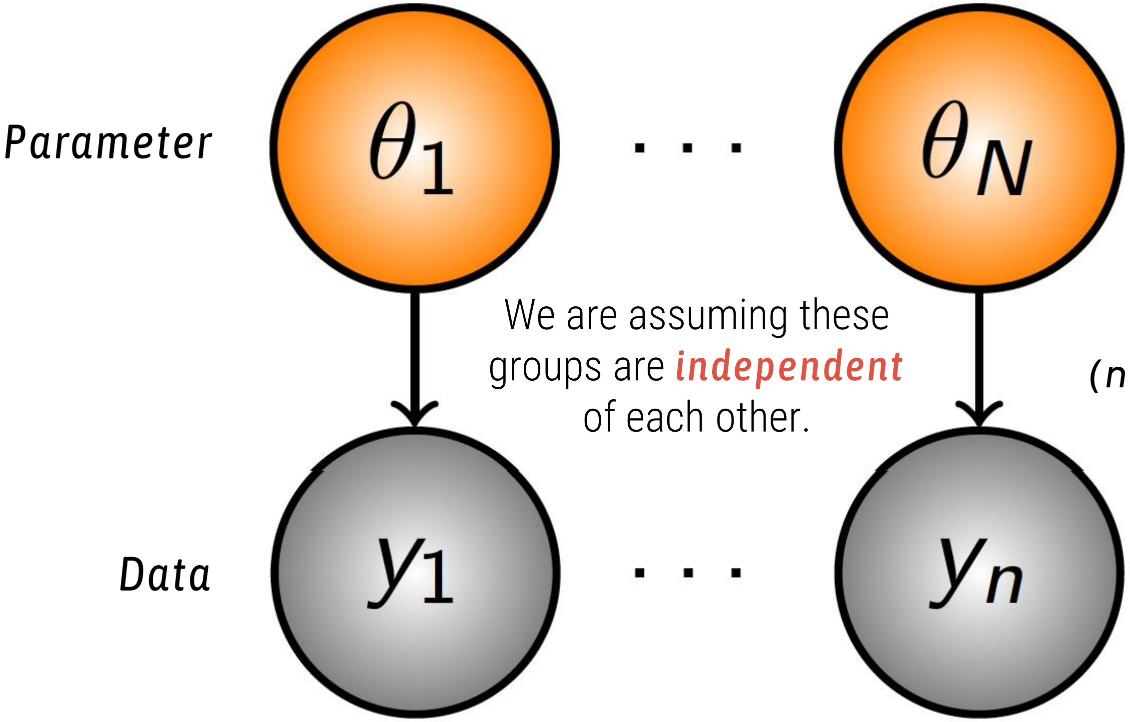
How does data inform parameters?



"No **pooling**"
(no sharing of information)

Adapted from Midway (2008)

HIERARCHICAL MODELS

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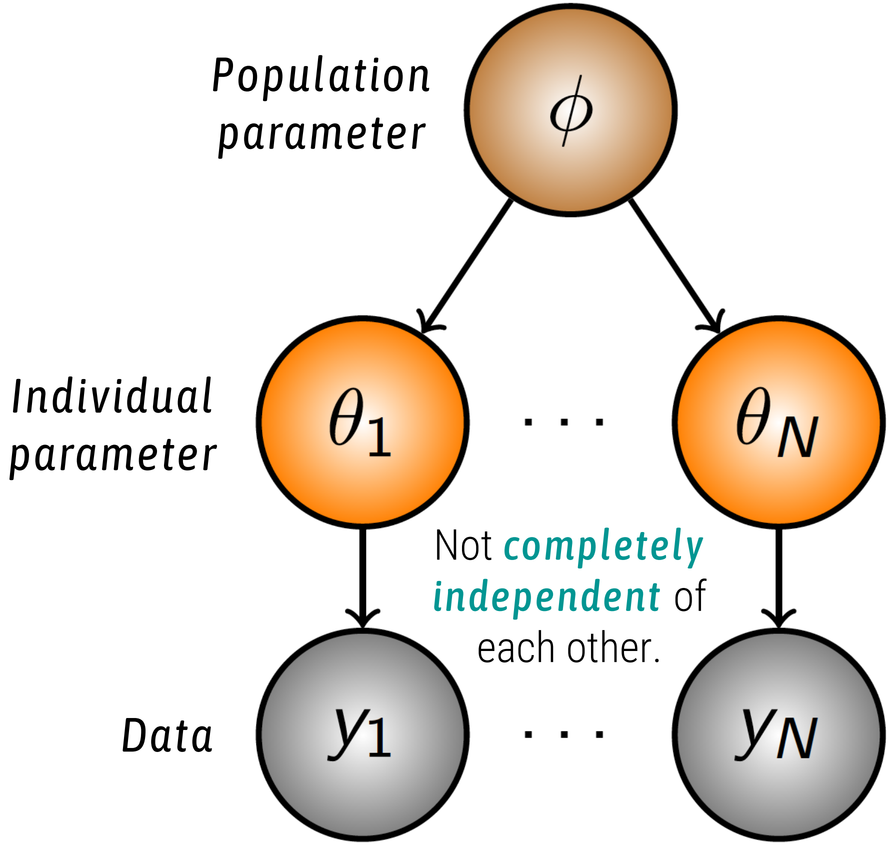

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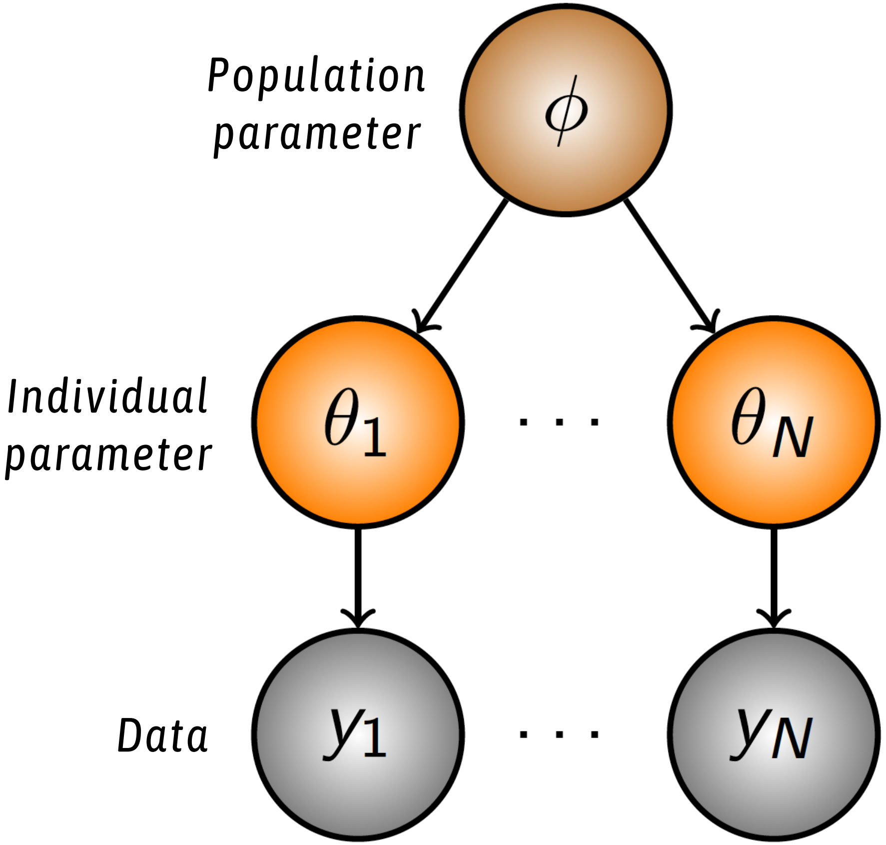
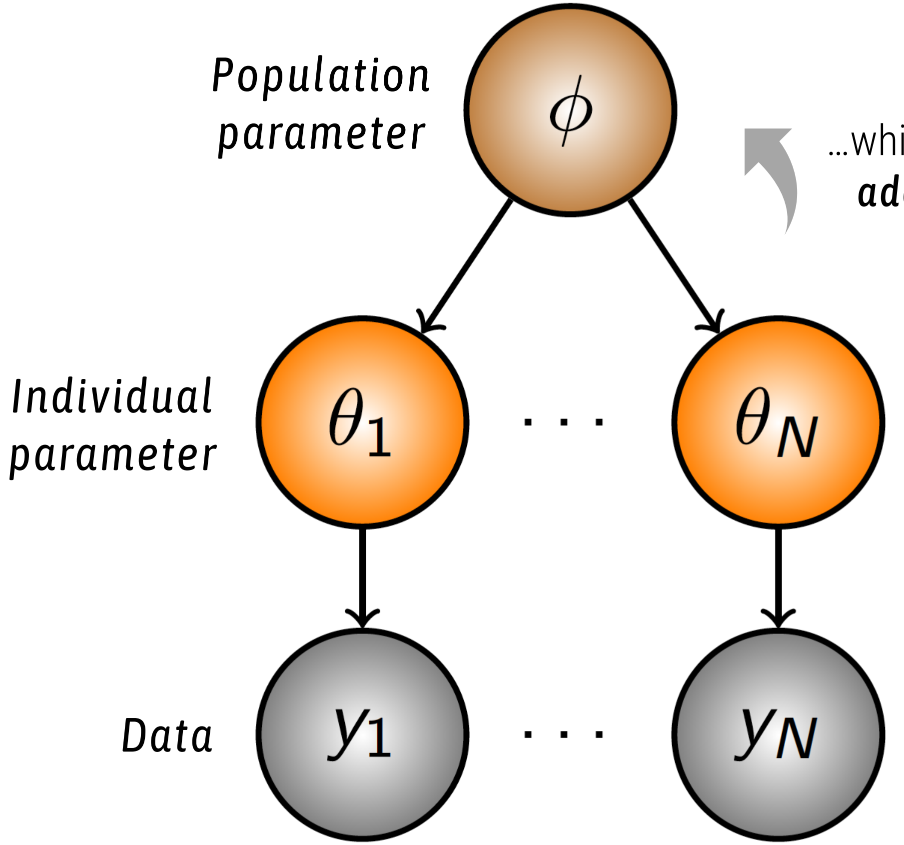


Data inform ***factor-level parameters***, θ…

Adapted from Midway (2008)

HIERARCHICAL MODELS

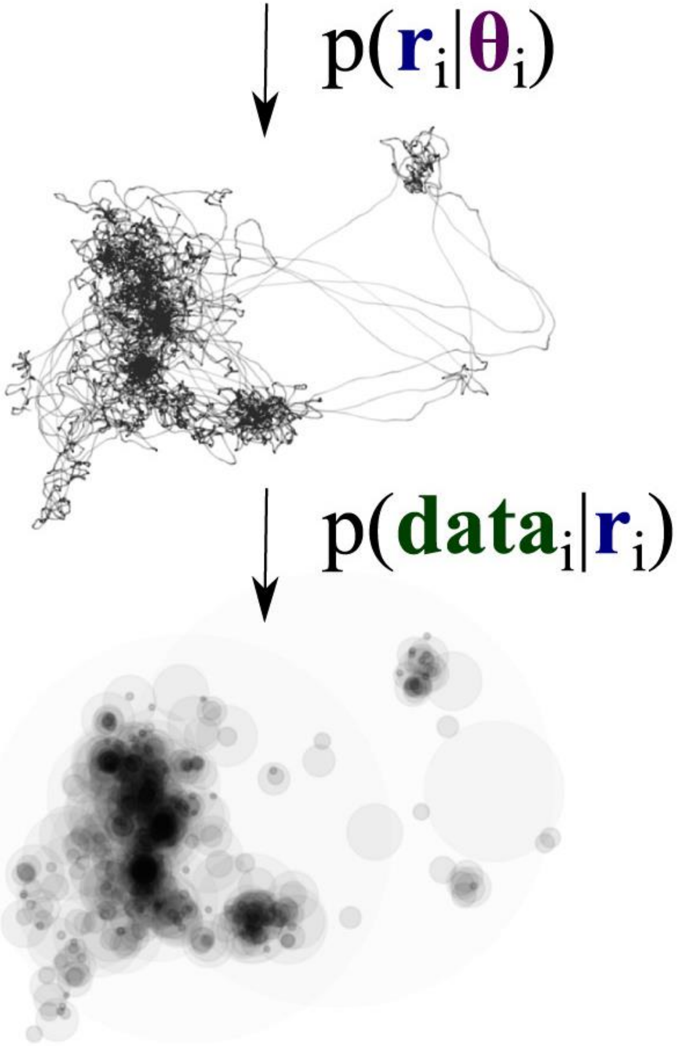
How does data inform parameters?



...which are governed by some ***additional parameter***, Φ.

Data inform ***factor-level parameters***, θ...

Adapted from **Midway (2008)**



$$p(\mathbf{DATA}|\boldsymbol{\Theta}) = \prod_i^m \int d\mathbf{r}_i \int \boldsymbol{\theta}_i \underbrace{p(\mathbf{data}_i|\mathbf{r}_i)}_{\text{observation}} \underbrace{p(\mathbf{r}_i|\boldsymbol{\theta}_i)}_{\text{movement}} \underbrace{p(\boldsymbol{\theta}_i|\boldsymbol{\Theta})}_{\text{population}},$$

$\mathbf{DATA}$ = tracking data from m individuals,

$\mathbf{data}_i$ = tracking data from individual i,

$\mathbf{r}_i$ = trajectory of individual i,

$\boldsymbol{\theta}_i$ = movement characteristics (e.g., home range) of individual i,

$\boldsymbol{\Theta}$ = Population parameters (e.g., variation in home ranges).

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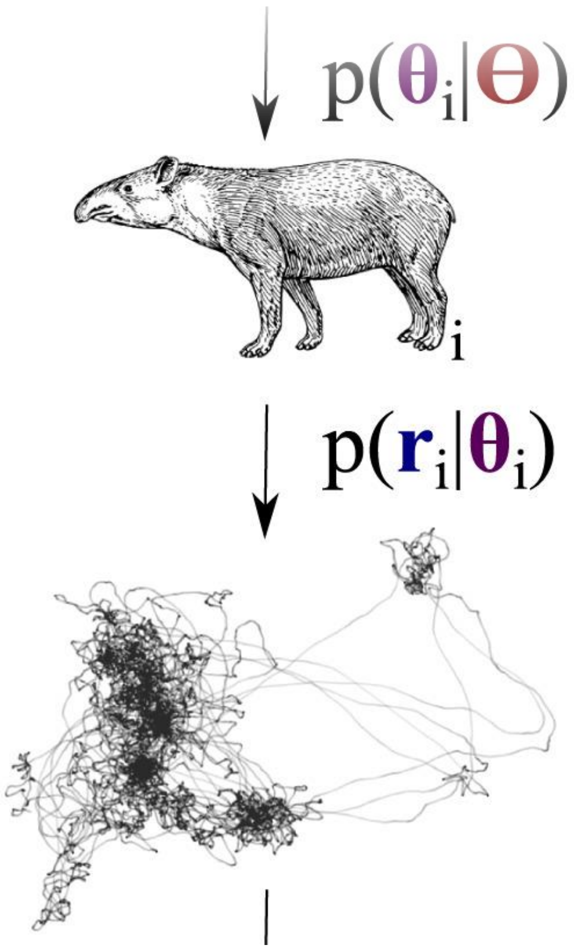

Individual

Trajectory

$$p(\mathbf{DATA}|\boldsymbol{\Theta}) = \prod_{i}^{m} \int d\mathbf{r}_i \int \boldsymbol{\theta}_i \underbrace{p(\mathbf{data}_i|\mathbf{r}_i)}_{\text{observation}} \underbrace{p(\mathbf{r}_i|\boldsymbol{\theta}_i)}_{\text{movement}} \underbrace{p(\boldsymbol{\theta}_i|\boldsymbol{\Theta})}_{\text{population}},$$

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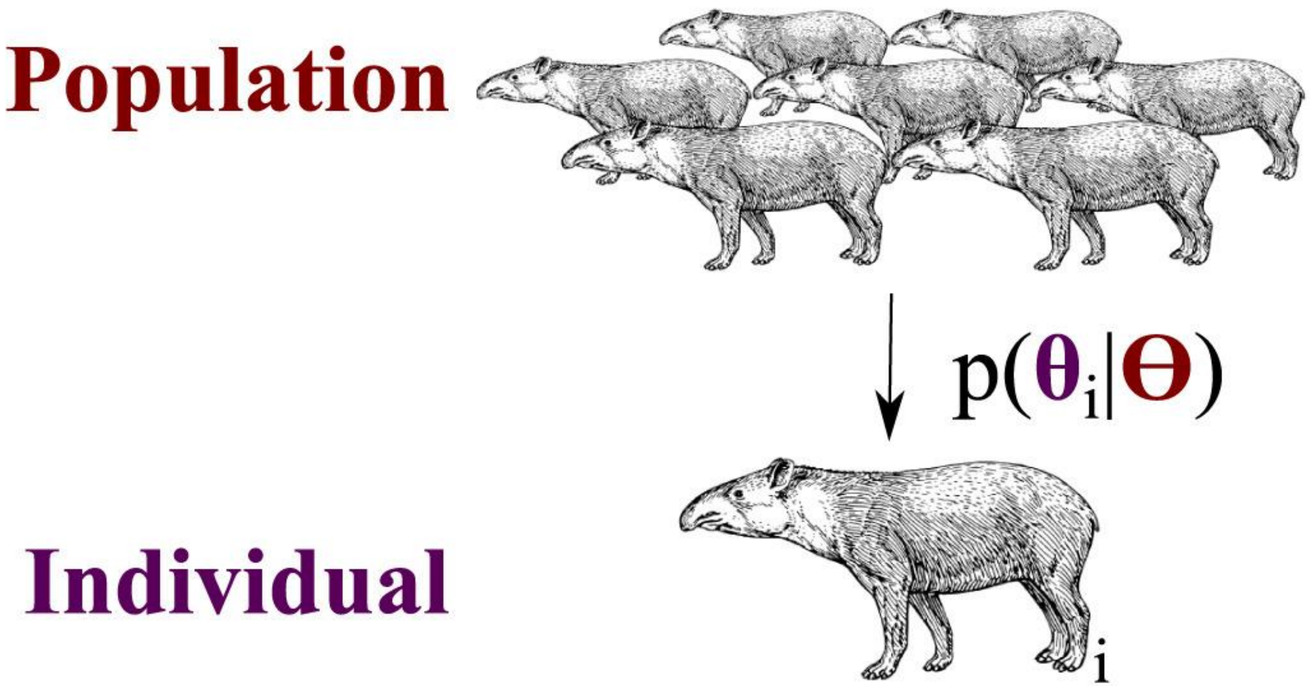
Population-level inferences



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χ^2 *inverse-Gaussian* (χ^2-IG) meta-analyses

$\mathcal{O}(1/N)$ bias
$\mathcal{O}(1/m)$ bias

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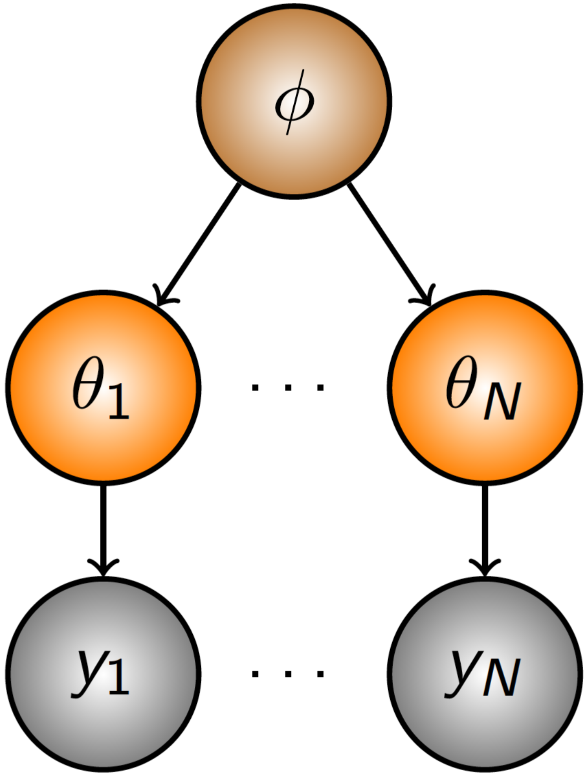
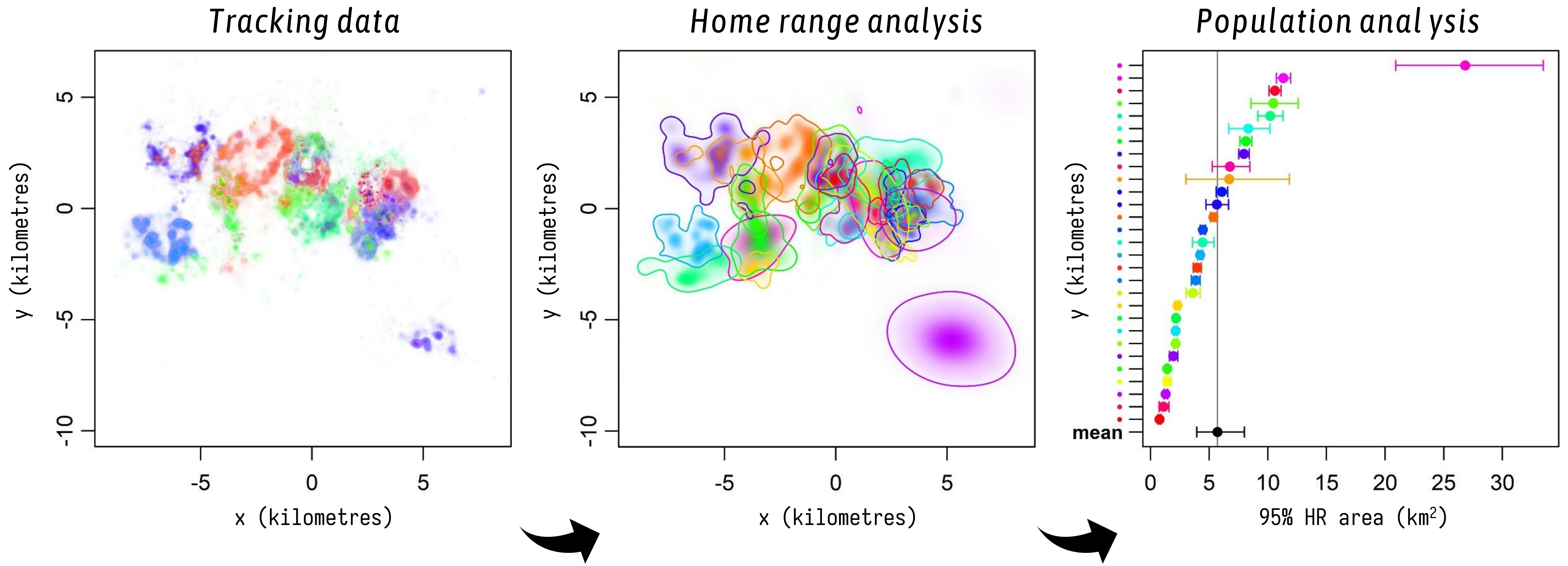


Account for the uncertainties in individual home-range estimates by treating the home-range areas as **unknown latent variables within a hierarchical model**.



Home range area estimates follow a χ^2 **sampling distribution**

Population of home-range areas follows an **inverse-Gaussian distribution**

A statistically efficient estimator will ***downweight*** uncertain estimates relative to more certain estimates in such a way that the estimated mean has a smaller variance.

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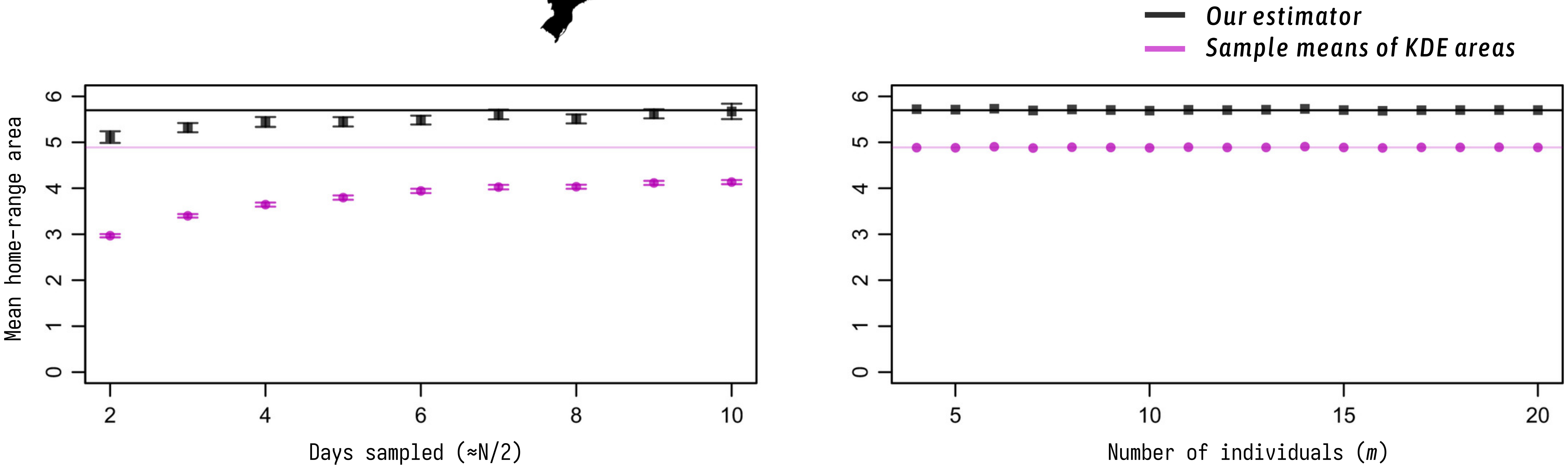
Lowland tapir

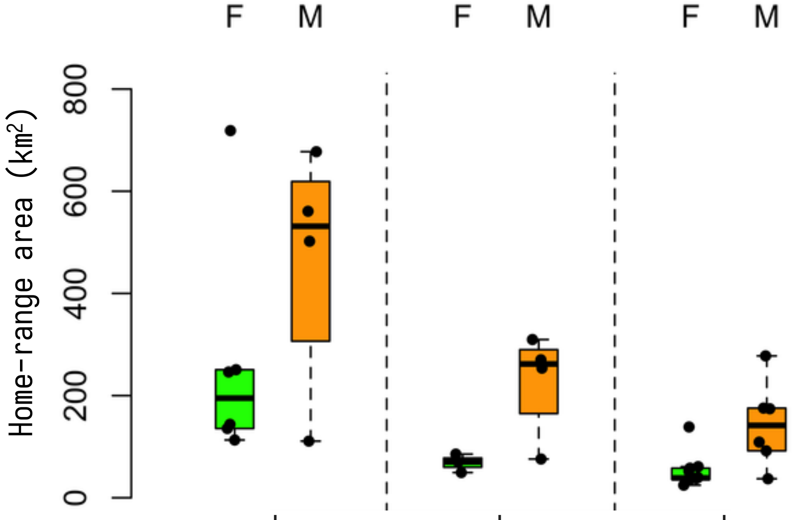
(*Tapirus terrestris*)

Tapirs have ***home range crossing times*** (τ_p) of 0.72 days, (ranging from 0.05–12.8 days)



Morato *et al.* (2016)

DOI: 10.1093/jmammal/gyab068



Jaguar

(Panthera onca)

Pete Wilcox

Atlantic forest

Amazon

Pantanal

Frans Lanting

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`ctmm_akde.R`

Critical patch size

Minimum area requirements

Theoretical concepts

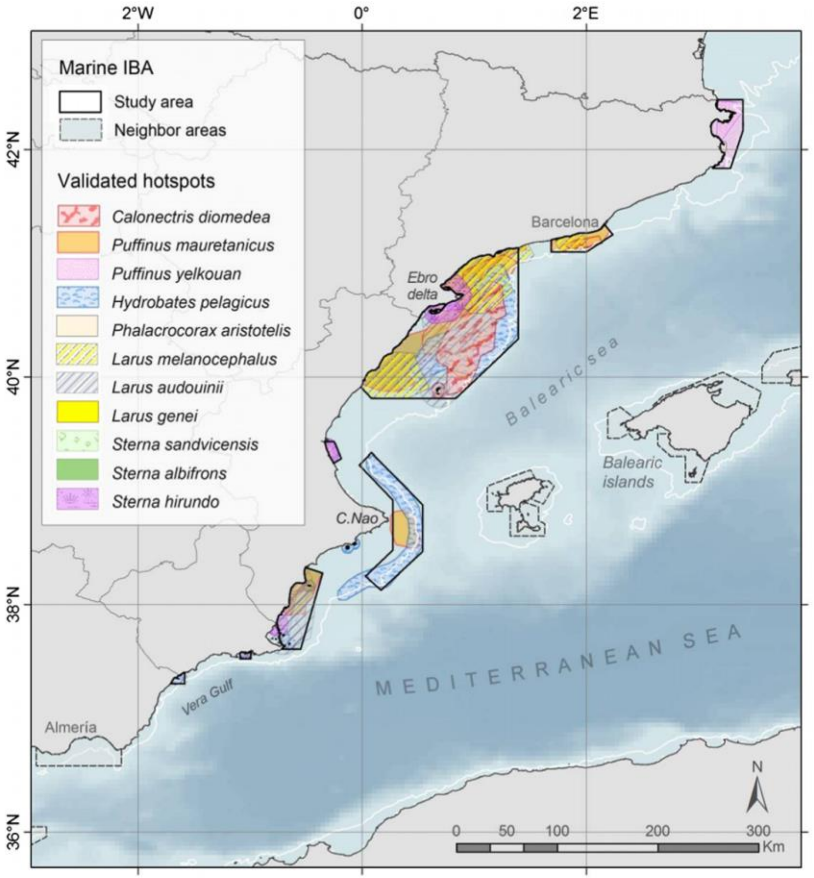
Population-level inferences

Real work applications

Conservation

Management

Using foraging hotspots of pelagic seabirds to identify marine **Important Bird Areas (IBAs)** in Spain.



Arcos *et al.* (2012)

DOI: 10.1016/j.biocon.2011.12.011

What's the mean home range area?

Average area used by individuals in a sample

What's the population distribution?

Spatial extent of the population as a whole

Methods:

- (A)KDE of population?
- Union of (A)KDEs?
- Mean of (A)KDEs?

Dual challenge:

Individual temporal autocorrelation
Population variation

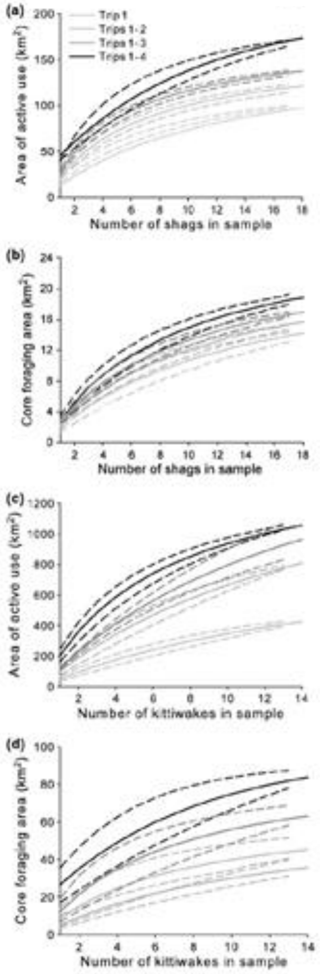
Sample
Tracked individuals

≠

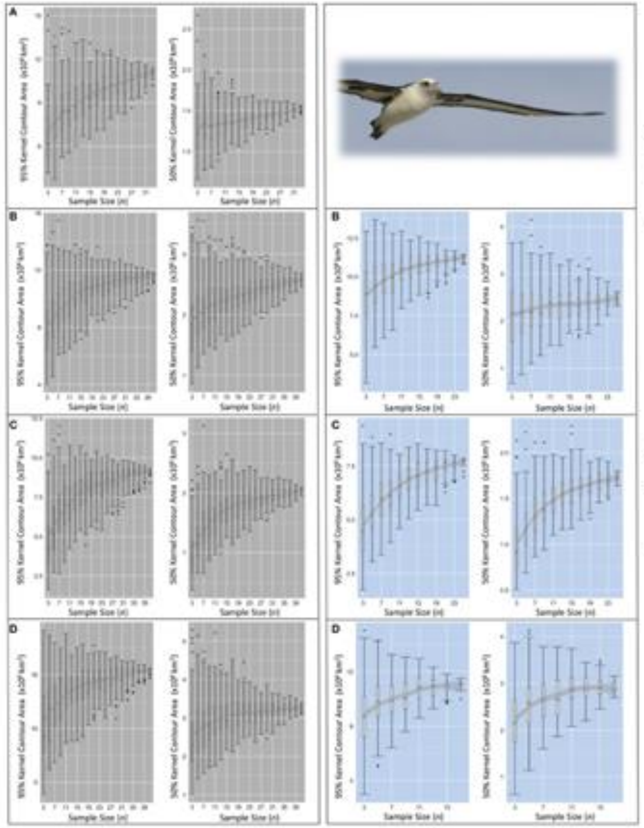
Population
Tracked + untracked individuals

Sample size vs. saturation

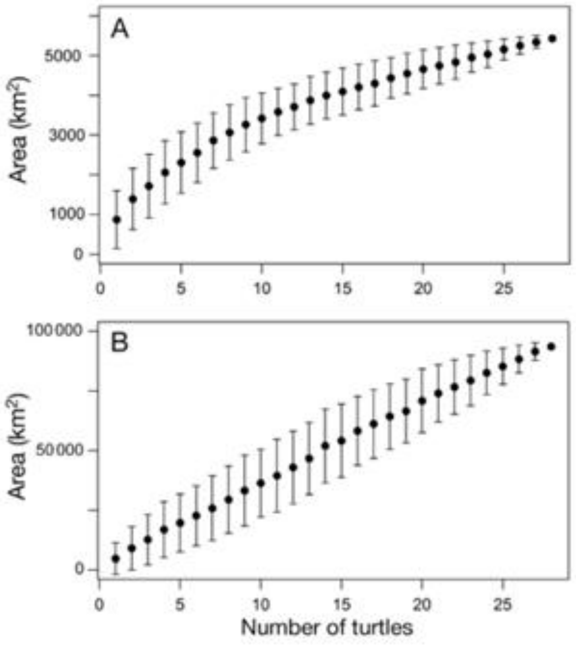
- Extrapolation of asymptote



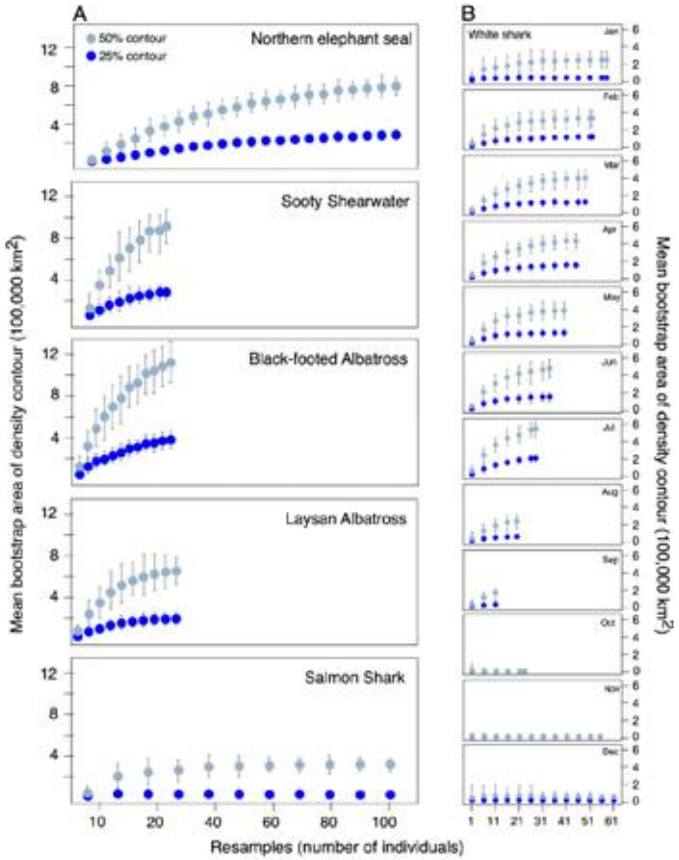
Soanes et al. (2013)



Gutowsky et al. (2015)



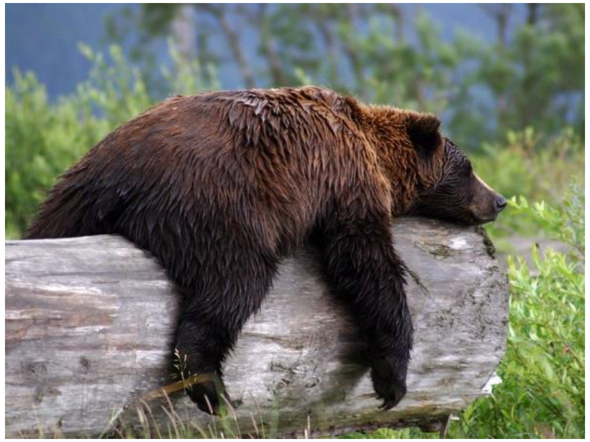
Thums et al. (2018)



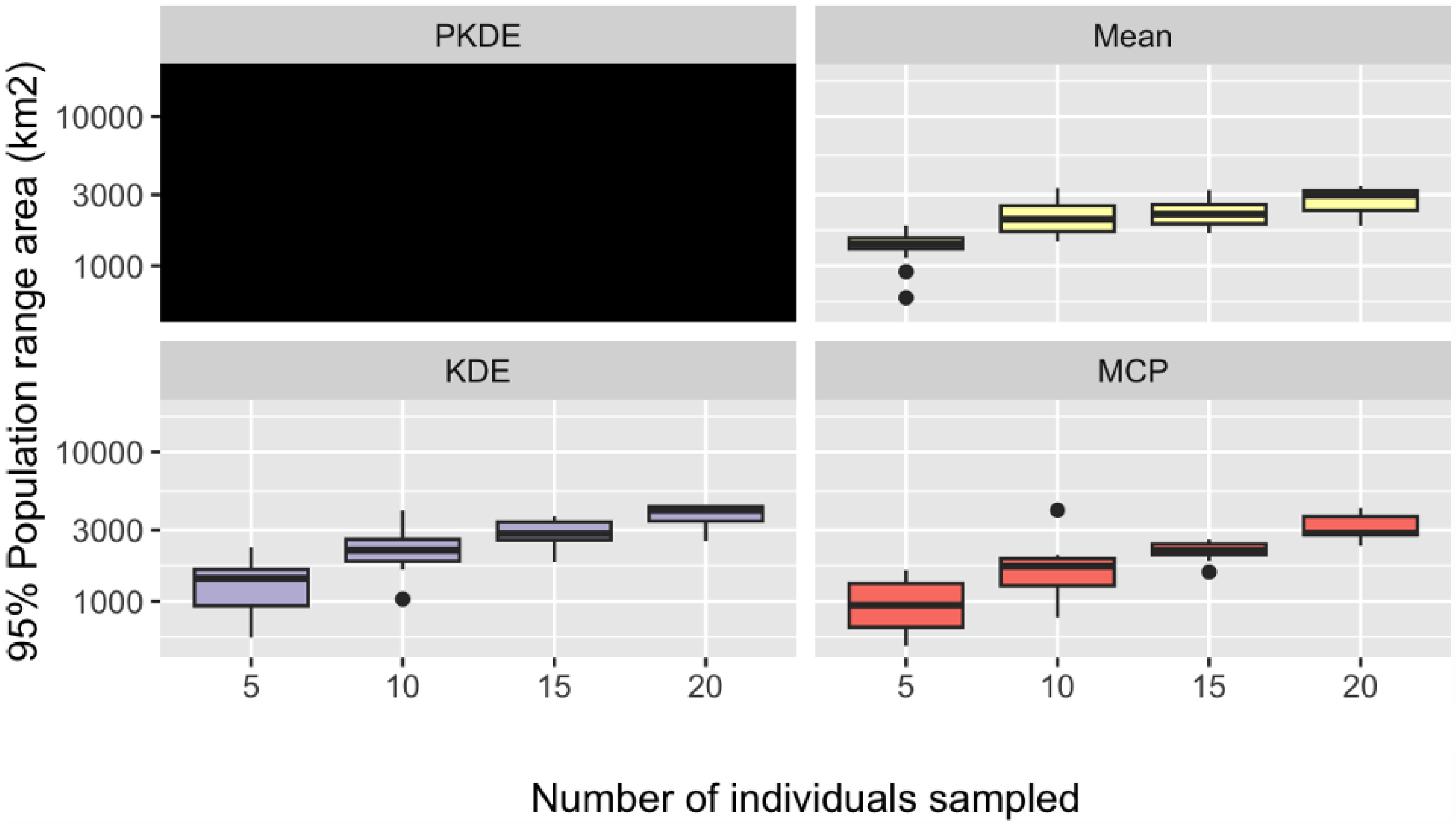
Sequeira et al. (2019)

- Extrapolation of asymptote

Saturation curve



Grizzly Bear
(Ursus arctos horribilis)



This bias is from *not* modeling population variance!

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A KDE is a weighted average of kernels:

$$\hat{p}(\mathbf{x}|\mathbf{H}) = \sum_{t}^{n} w(t)\,\kappa(\mathbf{x}-\mathbf{x}(t)|\mathbf{H})$$

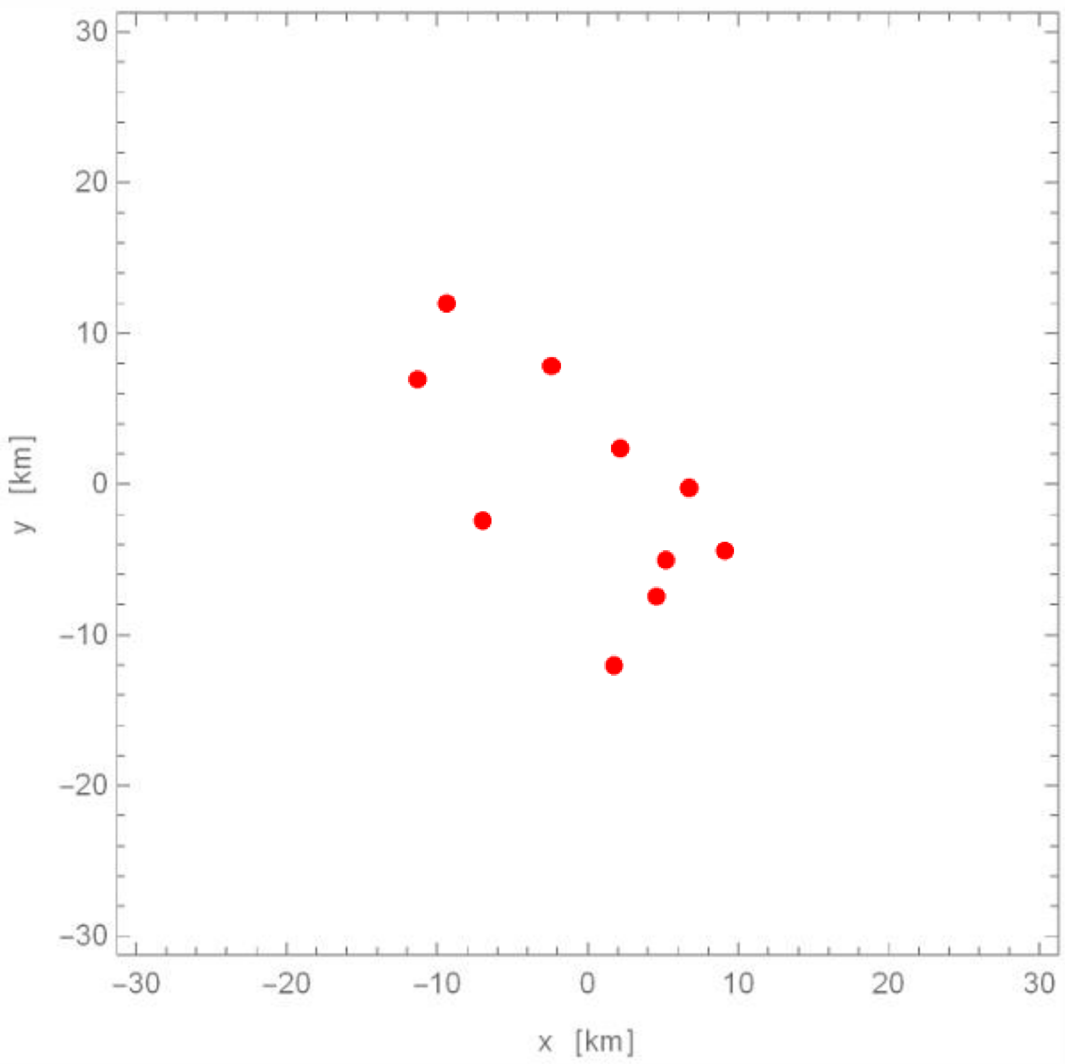
$\hat{p}(\cdot)$: probability density estimate

$\mathbf{x}$: location of interest

$w(t)$: weight at time t

$\kappa(\cdot)$: kernel function (e.g., Gaussian)

$\mathbf{x}(t)$: sampled location at time t



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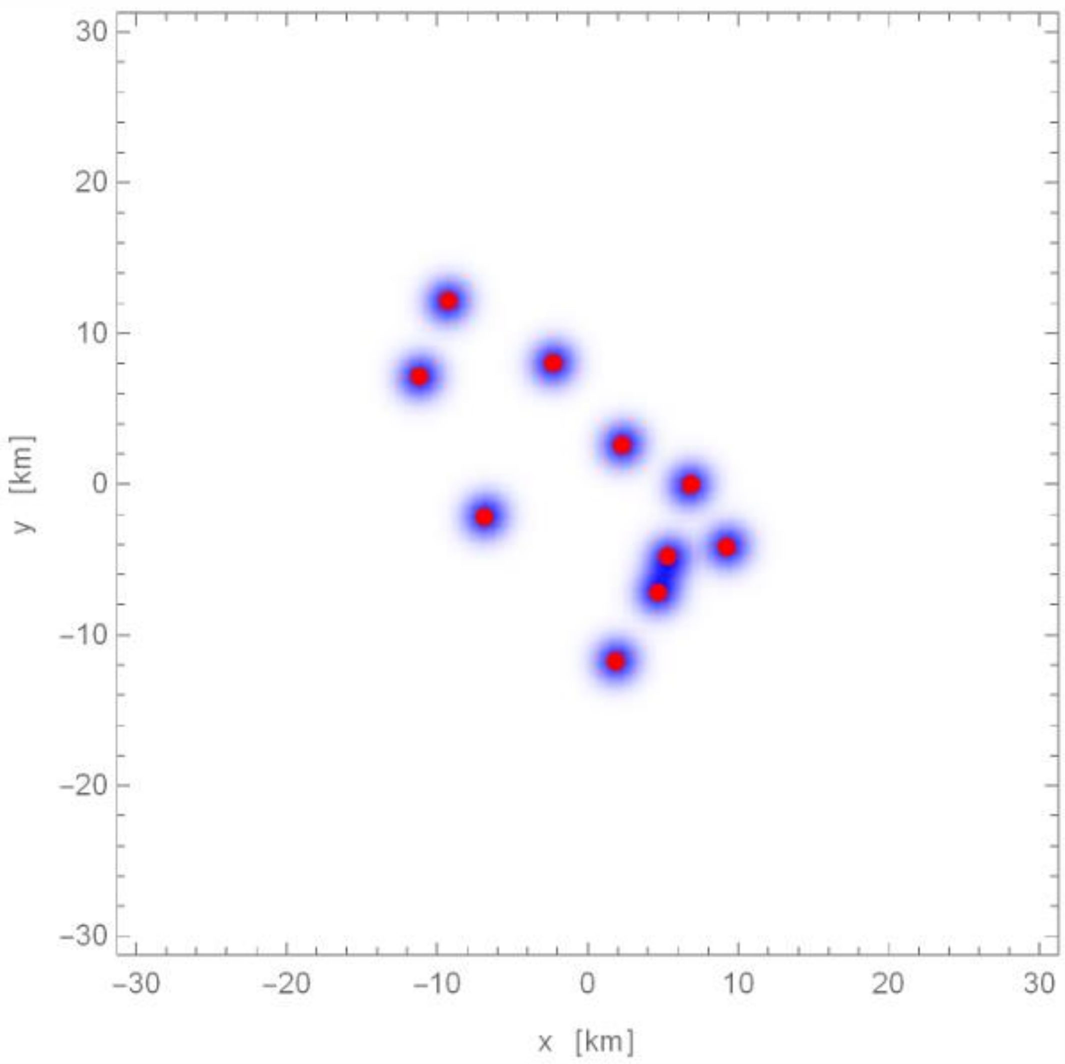
$\mathbf{H}$: bandwidth (e.g., COV of kernel, κ)

Where the optimal $\mathbf{H}$ minimizes the MISE:

$$\mathrm{MISE}[\mathbf{H}] = \mathrm{E}\left[\iint (\hat{p}(\mathbf{x}|\mathbf{H}) - p(\mathbf{x}))^2 d\mathbf{x}\right]$$



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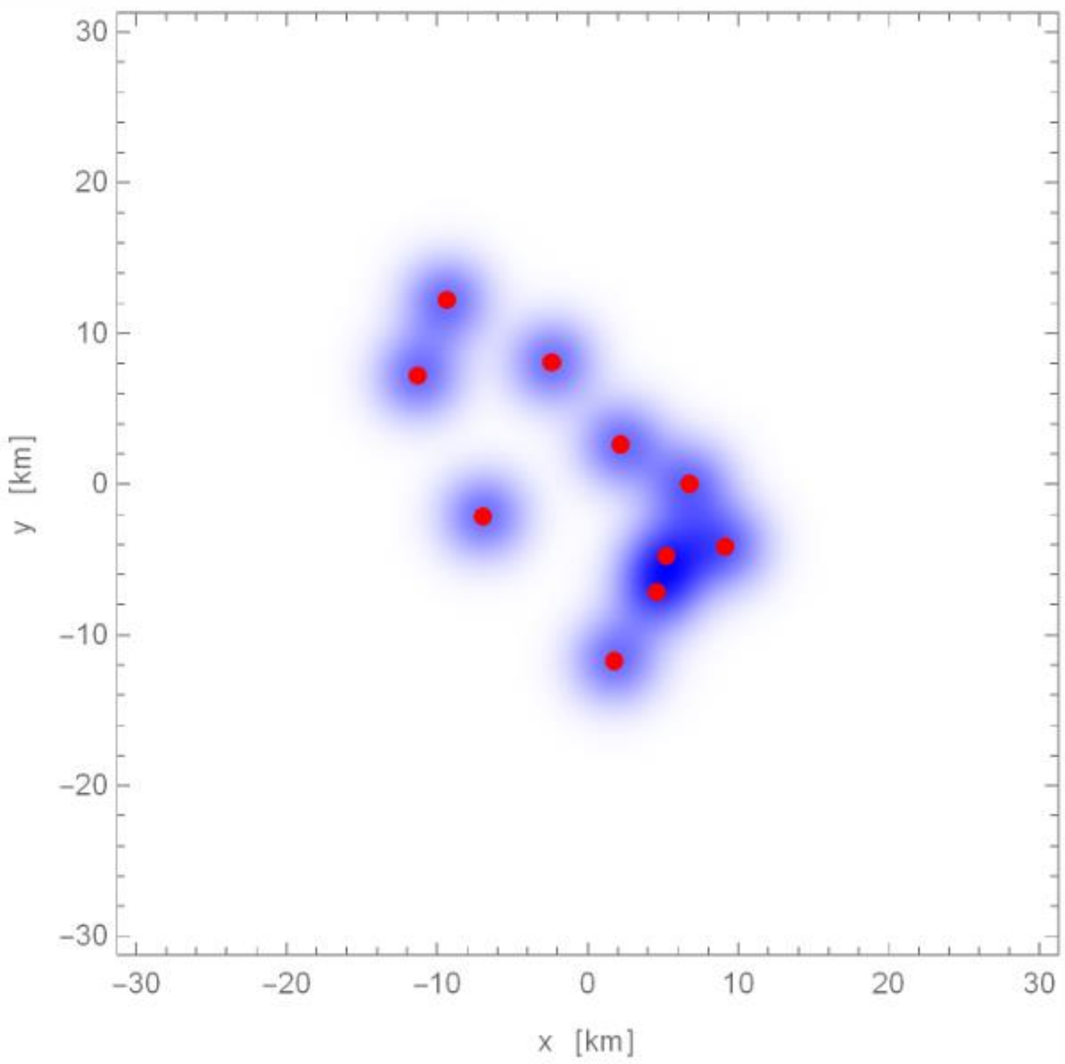

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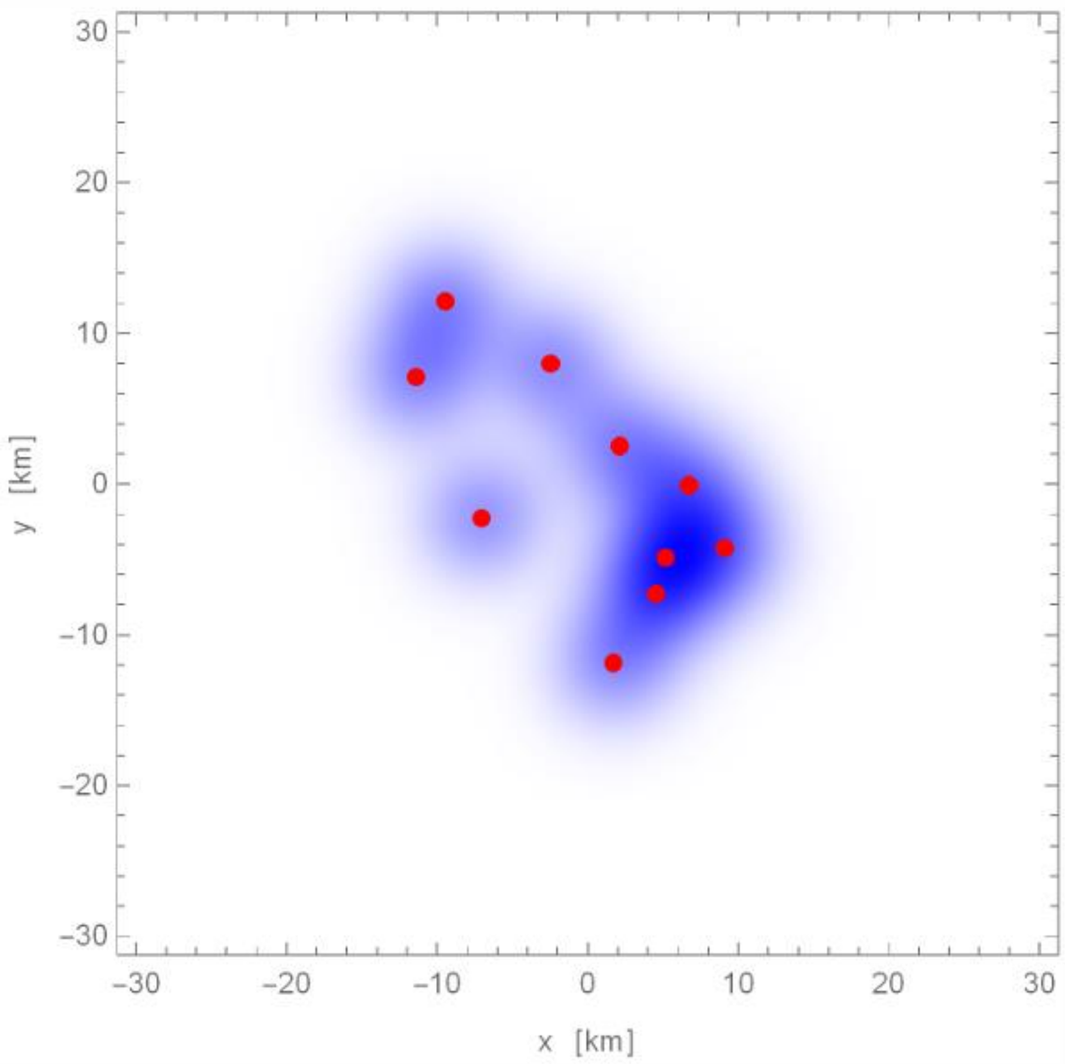

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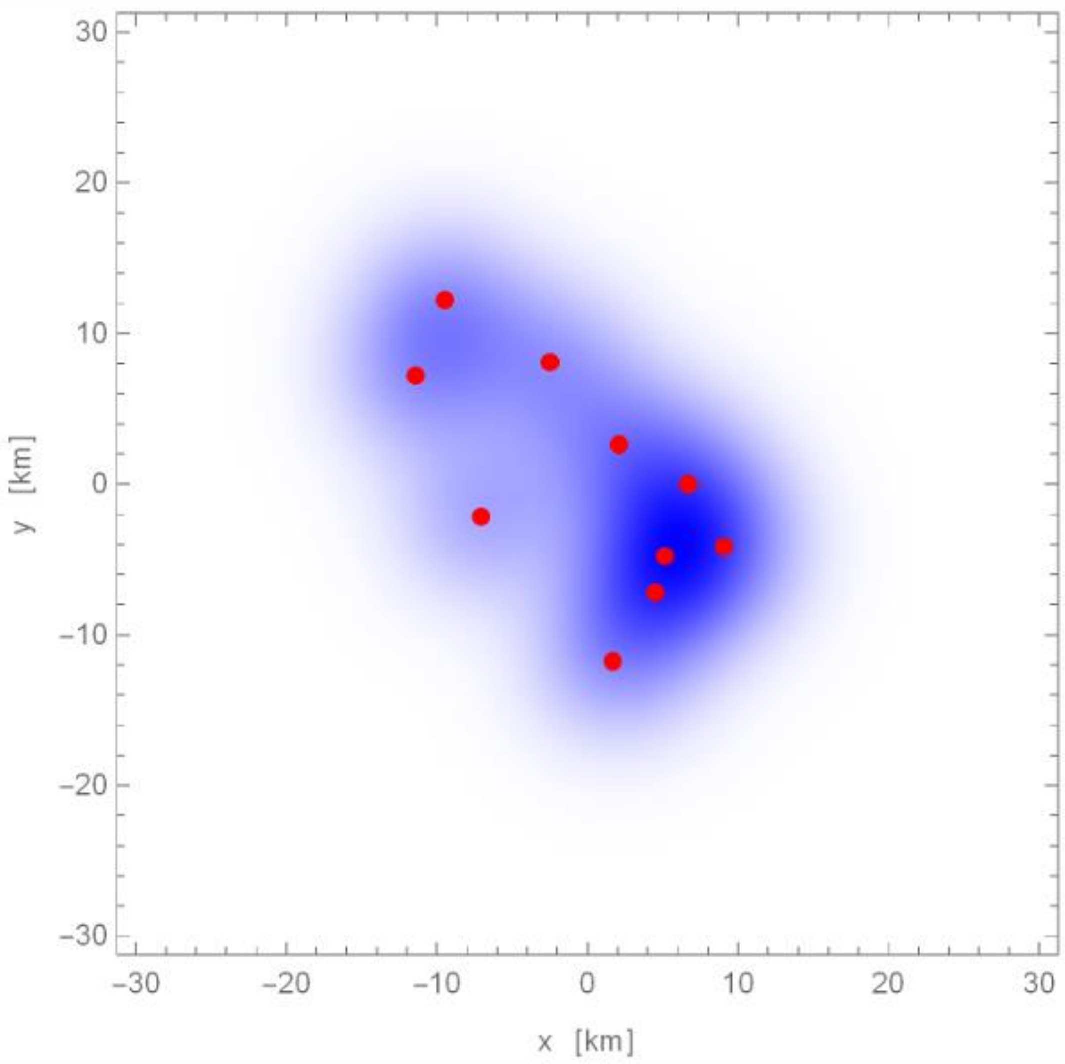

$$\mathrm{MISE}[\mathbf{H}] = \mathrm{E}\left[\iint \left(\hat{p}(\mathbf{x}|\mathbf{H}) - p(\mathbf{x})\right)^2 d\mathbf{x}\right]$$

$$p(\mathbf{x}) = \text{approximation (e.g., Gaussian reference function)}$$

$$\mathbf{x}(t) \sim \text{IID (conventional KDE)}$$

$$\mathbf{x}(t) \sim \text{ACF (autocorrelated KDE)}$$

- Bandwidth optimization (Turlach, 1993)
- AKDE (Fleming *et al.*, 2015)
- Bias-corrected AKDEc (Fleming & Calabrese, 2017)
- (MISE) optimally weighted wAKDE (Fleming et al., 2018)

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$$\mathrm{MISE}[\mathbf{H}] = \mathrm{E}\left[\iint \left(\hat{p}_{\mathrm{pop.}}(\mathbf{x}|\mathbf{H}) - p_{\mathrm{pop.}}(\mathbf{x})\right)^2 d\mathbf{x}\right]$$

$$\mathrm{MISE}[\mathbf{H}] = \mathrm{E}\left[\iint \left(\hat{p}_{\mathrm{pop.}}(\mathbf{x}|\mathbf{H}) - p_{\mathrm{pop.}}(\mathbf{x})\right)^2 d\mathbf{x}\right]$$

$$\hat{p}_{\mathrm{pop.}}(\mathbf{x}|\mathbf{H}) = \sum_{\mathrm{ind.}} W_{\mathrm{ind.}} \sum_{t_{\mathrm{ind.}}} w_{\mathrm{ind.}}(t_{\mathrm{ind.}})\, \kappa(\mathbf{x} - \mathbf{x}_{\mathrm{ind.}}(t_{\mathrm{ind.}})|\mathbf{H}_{\mathrm{ind.}})$$

$p_{\mathrm{pop}}(\mathrm{x})$ requires a hierarchical approximation

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$p_{\mathrm{pop}}(\mathrm{x})$ requires a hierarchical approximation

Individual weight assumptions:

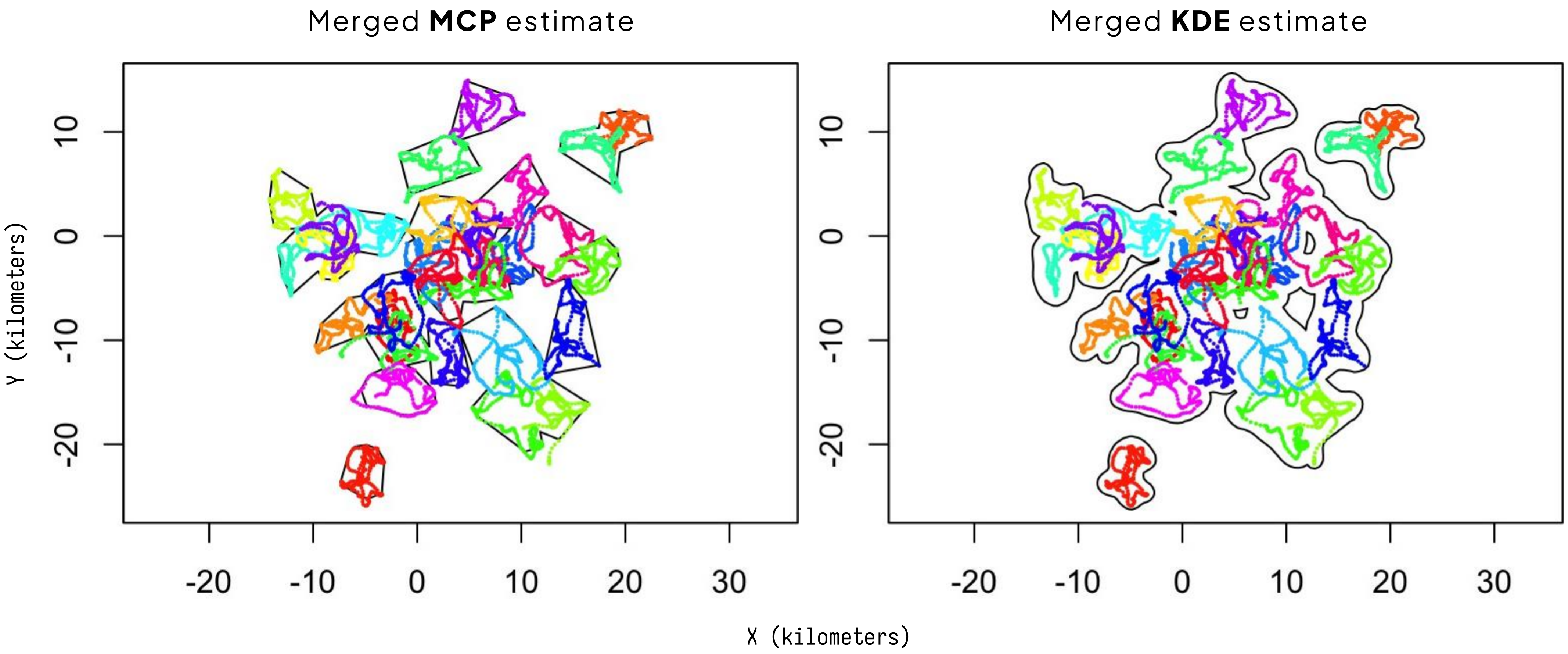
Uniform: $\sum_t w_{\mathrm{ind.}}(t) = \frac{1}{\sum_{\mathrm{ind.}}}$

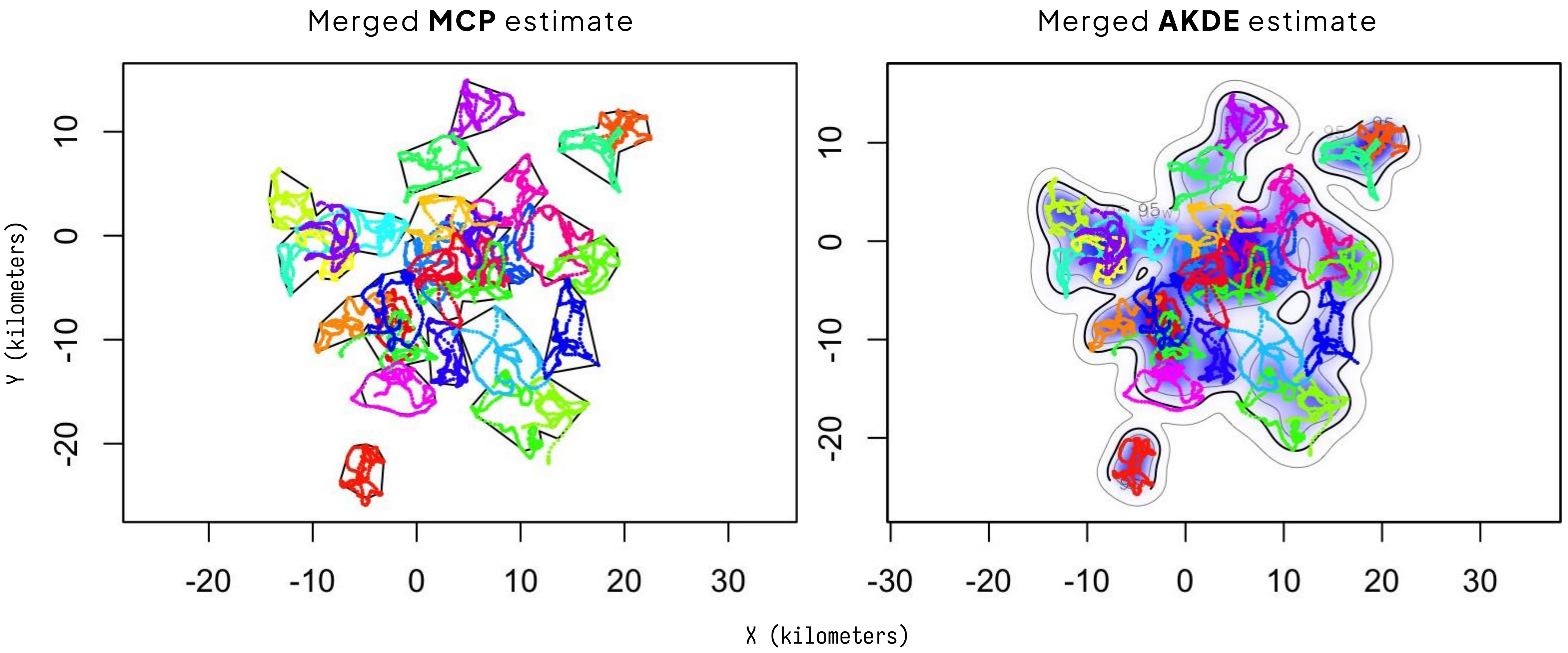
MISE-optimized: $\sum_t w_{\mathrm{ind.}}(t)$ (tends to zero-weight overlapped individuals)

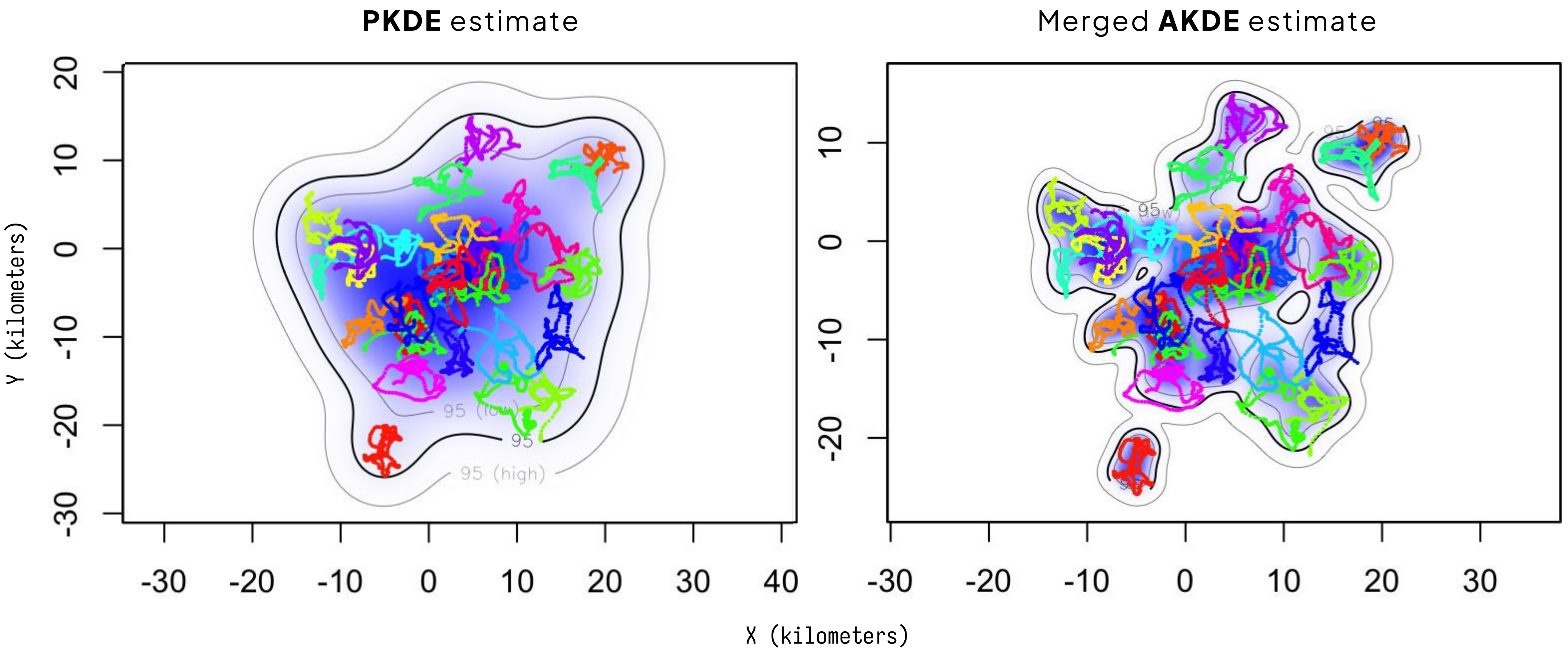
Bandwidth assumptions:

Proportional to individual: $\mathbf{H}_{\mathrm{ind.}} = h^2\, \mathrm{COV}[\mathbf{x}_{\mathrm{ind.}}]$ (fast)

Proportional to population: $\mathbf{H}_{\mathrm{ind.}} = h^2\, \mathrm{COV}[\mathbf{x}_{\mathrm{pop.}}]$ (slow)


Merged **MCP** estimate
Merged **KDE** estimate
10
0
-10
-20
Y (kilometers)
-20
-10
0
10
20
30
X (kilometers)


Merged **MCP** estimate
Merged **AKDE** estimate
10
0
-10
-20
-20 -10 0 10 20 30
-30 -20 -10 0 10 20 30
Y (kilometers)
X (kilometers)



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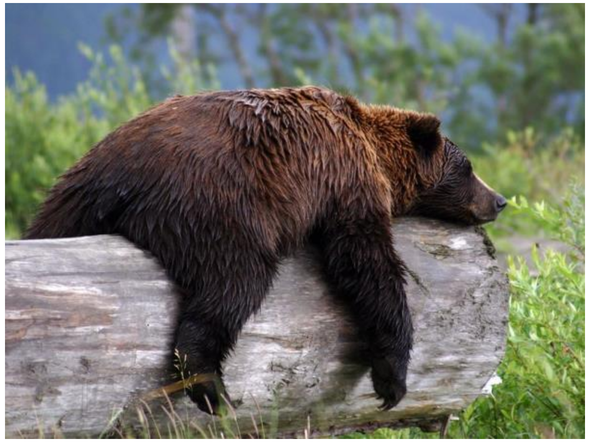
Sample size vs. saturation

– Extrapolation of asymptote



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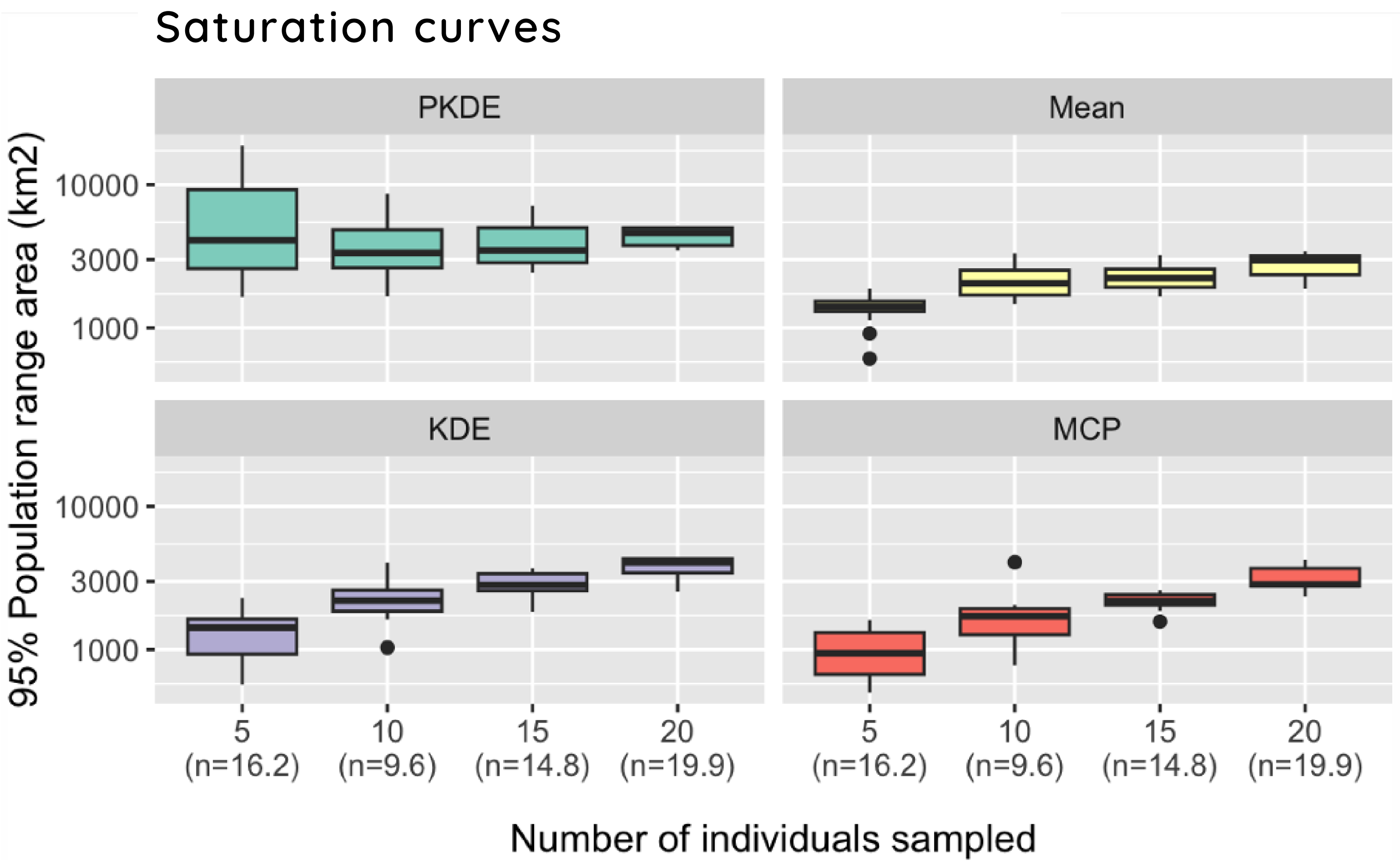

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PKDE solves the problem of how many individuals you need to sample

PKDE estimates saturate almost immediately, with a shrinking variance/uncertainty

Warning:

- If saturation curves reflect bias, decay is power law (slow) not exponential (fast)

Future directions:

- Downweighing poorly sampled individuals: $W_{\text{ind.}} \propto \left(1 + \frac{1}{\text{DOF}[\text{Area}_{\text{ind.}}]}\right)^{-1}$
- Downweighing non-representative individuals: $W_{\text{ind.}} \propto \langle p_{\text{ind.}} | p_{\text{pop.}} \rangle$

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